

Section 85 — Weighted And General Matching

Augmenting needs either a dual certificate or a contraction

This lesson develops A–C. Problem D is editorial-only.

Problem A: Hungarian potentials

Assignment is perfect bipartite matching. Hungarian maintains row potentials u , column potentials v , and nonnegative reduced costs $\text{cost} - u - v$. Equality edges have zero reduced cost. While inserting one row, grow an alternating tree; the smallest slack changes potentials until a new equality edge appears. A free column ends an augmenting predecessor chain.

Weak duality makes $\text{sum}(u) + \text{sum}(v)$ a lower bound. A perfect matching of equality edges reaches that bound, so complementary slackness proves optimality.

Problem B: forbidden rectangular assignment

The same algorithm supports rows no more numerous than columns. Negate profit and replace forbidden pairs by an infinity larger than every feasible total. After optimization, explicitly reject a matching containing infinity; otherwise its negated cost is the maximum profit.

Problem C: blossoms

In general graphs an edge between two even alternating-tree levels creates an odd alternating cycle. Contract that blossom: from outside, all alternating ways through it are interchangeable for finding an augmenting path. Continue BFS on bases, then use stored predecessor information to expand and flip the original path.

The invariant is that augmenting paths exist in the contracted graph exactly when they exist in the original graph.

A concrete Hungarian trace

Consider the cost matrix

	c0	c1	c2
r0	4	1	3
r1	2	0	5
r2	3	2	2

When a row is inserted, columns already reached by the alternating tree have known slack: the cheapest reduced-cost edge entering each column. If no free column is tight, subtract the minimum slack δ from every unreached column's slack and update the two sides' potentials. At least one new equality edge appears. Thus an iteration either enlarges the tree or finishes an augmentation; it never changes which complete assignment is optimal.

The useful mental model is not “Hungarian greedily chooses a cheap edge.” It maintains two objects at once:

1. a partial primal solution, the matching;
2. a dual certificate, the potentials.

The matching becomes optimal precisely when every selected edge is tight and all rows are matched.

Why an odd cycle must be contracted

In a bipartite alternating tree, parity identifies the two sides. In a general graph, an edge joining two even vertices closes an odd cycle. Trying to assign fixed even/odd labels inside that cycle now gives a contradiction. Contraction does not discard the choices: it postpones the decision of which alternating route through the cycle is used until an augmenting path enters and leaves the contracted base.

For a triangle with one vertex matched to the outside, no individual internal edge can safely be committed to. As one super-vertex, however, the triangle has exactly the interface relevant to the continuing BFS. Expansion later rotates its matching so that the discovered path becomes alternating.

Proof and implementation checklist

- For Hungarian, state dual feasibility, matching feasibility, and tightness of selected edges separately.
- Choose the forbidden-edge sentinel from a proved bound on every finite total; do not rely on a convenient-looking constant.
- In blossom code, distinguish an original vertex from its current base.
- Reset BFS queues and predecessor data for every augmentation, but preserve the current matching.
- Test empty graphs, triangles, nested blossoms, rectangular matrices, negative profits, and instances in which every completion uses a forbidden edge.

Exercises

A–C cover assignment duality and blossom contraction. D is a modeling transfer.